

Experimental apparatus for universal spin-motion control in multi-ion registers



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Contents

1	Towards Hybrid System Control and Quantum Error Mitigation	1
1.1	Continuous-Variable Control in Trapped Ions	2
1.1.1	Theory	3
	Computing on a Hybrid System	3
	Gate Implementation	4
	Readout	5
1.1.2	Experimental Demonstration	5
	Two Spin Dependent Forces	6
	Squeezing	6
1.1.3	Outlook	7
1.2	Virtual Distillation	7
1.2.1	Error Mitigation	7
1.2.2	Virtual Distillation Protocol	9
	Purification	9
	Circuit	12
1.2.3	Experimental Demonstration	13
	Implementation	13
	B-matrix Calibration	13
	Injecting Spatial Correlations	13
1.3	Proposed experiments and technical requirements	14
	Extending to > 2 copies	14
	Virtual Distillation on Hybrid Systems	15
1.4	Outlook	16
Appendices		
References		18

1

Towards Hybrid System Control and Quantum Error Mitigation

Contents

1.1	Continuous-Variable Control in Trapped Ions	2
1.1.1	Theory	3
1.1.2	Experimental Demonstration	5
1.1.3	Outlook	7
1.2	Virtual Distillation	7
1.2.1	Error Mitigation	7
1.2.2	Virtual Distillation Protocol	9
1.2.3	Experimental Demonstration	13
1.3	Proposed experiments and technical requirements . .	14
1.4	Outlook	16

Keep in mind: Want to make clear that this is future work, and not specifically QEM on CV!

Having established and characterised the experimental platform and demonstrated its application to *k-copy* randomised benchmarking, this chapter considers future research goals utilising the multi-qubit arrays enabled through the *NPL* trap (section ??), the single addressing system (section ??), and the developed control over both the discrete qubits, as well as the continuous motional modes. Following on from recent progress in spin-conditioned bosonic operations demonstrated within

our group [bazavan_thesis, saner_thesis, shinbrough_applications_2026, 1, 2], we are interested in developing the required control to utilise multiple qubits and motional modes for computation and simulation.

In section 1.1 we establish a set of unitary operations required for universal continuous-variable control of the ion motion and discuss the extensions to hybrid spin-oscillator systems. We outline the method we use to generate such spin-oscillator interactions experimentally, and demonstrate the generation of a squeezed state on the new apparatus. We then outline the next steps required for demonstrating the remaining single- and two-mode interactions, and the possible experiments this will enable.

The second half of this chapter considers an error mitigation technique applicable to both discrete and continuous systems. Developing this, and associated methods on our trapped ion platform will allow substantial reductions of certain errors, with significantly less resources required than full error correction. We give a short introduction to the purpose of quantum error mitigation, followed by an outline of virtual distillation [3–5] — the specific method considered here. We provide an initial demonstrations of this protocol applied to a discrete-variable only system, and explore the performance under both local depolarising and correlated noise due to a common fluctuating control field (see section ?? for details).

We conclude this chapter with a discussion on the application of virtual distillation to hybrid systems, presenting a scalable method within reach of our apparatus.

1.1 Continuous-Variable Control in Trapped Ions

One paragraphs on what is CV, and why it is interesting. One paragraph on its conception in ITQC. - CV offers an alternative strategy for universal computation, utilising oscillators, termed *qumodes* [<empty citation>] **Nice to get original ref here**, as the fundamental quantum object, rather than qubits. qumodes have large Hilbert spaces... Photonic platform has explored this for a while [<empty citation>] **Photonic review citations**.

- Although qubit control is mature on trapped ion platform, there has only recently

been a spike of interest in developing continuous variable control on the collective motional modes of the ions [[empty citation](#)][Big citation dump here](#)

- Benefit of using trapped ions for CV is that we have access to CV unitaries via spin-oscillator coupling via laser field, have long coherence times.
- This highlights an extension, the trapped ion platform is an ideal hybrid system — we could store information in both qubits and qumodes, and using an over-complete gate set, we may efficiently decompose circuits to utilise both subsystems [6]. In the lab scale, this also offers the opportunity to access larger Hilbert spaces, without solving the problem of scaling the trapped ion system to QCCD type architectures (probably best left to companies) with orders of magnitude more qubits.
- The apparatus developed in this thesis is a good fit for exploring and scaling CVQC in trapped ions as we have access to:
 - Multiple qubits
 - Multiple modes
 - Good motional coherence times
 - Single addressing
 - We will have standing wave for generating strong qubit-oscillator couplings that are non-linear in ladder operators.

1.1.1 Theory

Computing on a Hybrid System

Summary from a couple of reviews (Liu?)

- Intro representations of a mode. Number and quadrature, mention phase space, wigner, others.
- For CV gateset need Gaussian and at least one non-Gaussian operation (c.f. Clifford and non-Clifford gates), Wigner negativity. See table III.1 in Liu.
- Define gaussian as at most quadratic in x and p .
- Freedom to choose the native operations, we present a particularly accessible set given light-matter interactions available on trapped ions. This is a cubic instruction

Interaction name	Oscillator H	Gauss.	N_{osc}	Two-SDF method		
				m	$\hat{\sigma}$	Ω
Displacement	$ae^{i\phi} + a^\dagger e^{-i\phi}$	✓	1	-	$\hat{\sigma}_x$	Ω_α
Parity	$aa^\dagger e^{i\phi} + a^\dagger a e^{-i\phi}$	✓	1	+1	$\hat{\sigma}_z$	Ω_2
Beam splitter	$ab^\dagger e^{i\phi} + a^\dagger b e^{-i\phi}$	✓	2	+1	$\hat{\sigma}_z$	Ω_2
Squeezing	$a^2 e^{i\phi} + a^{\dagger 2} e^{-i\phi}$	✓	1	-1	$\hat{\sigma}_z$	Ω_2
Two-mode squeezing	$abe^{i\phi} + a^\dagger b^\dagger e^{-i\phi}$	✓	2	-1	$\hat{\sigma}_z$	Ω_2
Trisqueezing	$a^3 e^{i\phi} + a^{\dagger 3} e^{-i\phi}$	✗	1	-2	$\hat{\sigma}_x$	Ω_3

Table 1.1: A suggested set of unitary interactions to compose a universal gate set for continuous variable computation. Using the protocol presented in reference [7], here termed the “Two-SDF” method, we show the required relative detuning setting, m , and the resulting spin conditioning, $\hat{\sigma}$, and coupling strength Ω of the resulting interaction. Ω_α is the typical SDF strength given by equation ??ref eqn in SDF theory sec. The other coupling strengths are found via the 2nd and 3rd terms of the Magnus expansion: $\Omega_2 = 2\Omega_\alpha\Omega_{\alpha'}/\Delta$, $\Omega_3 = 2\Omega_\alpha^2\Omega_{\alpha'}/\Delta^2$.

set (Can expand details from Liu III.E.2).

- Quick overview of the interactions and where we have demonstrated them. Could give flow fields for single-mode operators (could be a nice figure, and I like that view).

Gate Implementation

- Now how do we realise these interactions? - Present concept of Two-SDF toolkit from TS+RS
- Give Magnus expansion, and commutations (check SS and OB thesis theory sections)
- This then gives m , Δ , and Ω (point to table).
- Give short contrast with SB method.
- Here can also mention SW, cite recent ETH tweezer paper, and that we have even sharper gradient.

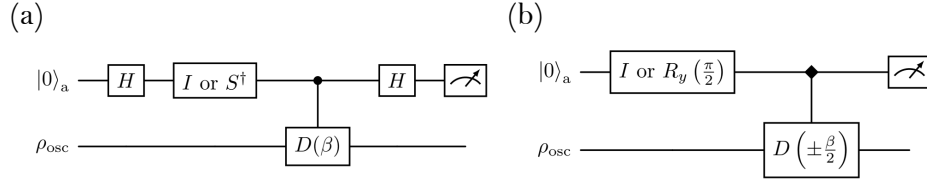


Figure 1.1: State tomography for a single mode through measurement of an ancilla qubit. (a) Displays the Hadamard test method for measuring the characteristic function, $\chi(\beta)$, of the oscillator using a displacement controlled by the qubit. The real and imaginary part of $\chi(\beta)$ is recovered through the application of either the identity, or the $S^\dagger$ gate before the displacement, respectively. (b) An equivalent measurement of $\chi(\beta)$ using native single qubit rotations and spin-dependent forces conditioned on σ_x (represented by the diamond control symbol).

Readout

- We don't have direct access to readout motion.
- So instead we map motional state to qubit.
- We can generally do this via Hadamard test (generalise to U in figure) for phase kickback on spin
- For full state tomography, we can measure characteristic function [[Fluhman, OB, SS thesis](#)]
- Give circuit for tomography using SDFs.
- Define Characteristic function.
- Can then compute Wigner or other representation of choice either through direct Fourier transform, or through ML methods (theorist we spoke to?)

$$\mathcal{P}_a = \frac{1 + \text{Re}(\xi(\beta))}{2}, \quad (1.1)$$

$$\mathcal{P}_a = \frac{1 + \text{Im}(\xi(\beta))}{2}, \quad (1.2)$$

1.1.2 Experimental Demonstration

Short description of what we will show - Frame as demo of squeezing with added complexity of using single addressing system - Point to motional coherence time. Was dephasing dominated, improved by grounding

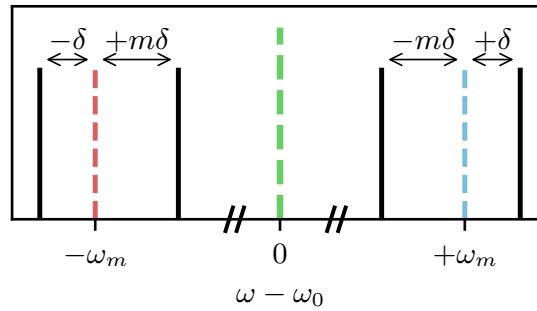


Figure 1.2: Figure: detuning for four tones, Z measurement with ramp The two bichromatic fields required for generating non-linear motional interactions via the two-SDF method. Motional mode frequency is given by ω_m , the first SDF is detuned by δ , and the second is detuned by $m\delta$, with the desired interaction selected by varying m . **change symbol m, and swap side of 2nd SDF.**

- Demonstrate squeezing of motion conditioned on spin.
- One ion, one motional mode.
- Using single addressing system

Two Spin Dependent Forces

- Experimental sequence, state prep, doppler, SBC, ramped 4 tones, SDF.
- Four tones on one AOM. How we set and calibrate phases. Maybe noise sidebands on AOM due to non-linearity. Balancing power.

Squeezing

- Detuning and power selected (only 40 uW!), and predicted strength of interaction?
- Check if ramp is sufficient by looking at σ_z dependence.
- Calibrate probe SDF and get strength.
- Calculate expected squeezing strength.
- Show char function of squeezed state.
- Show r vs duration, with detail of the ramp duration being accounted for.
- Discuss observed strength vs expected strength.

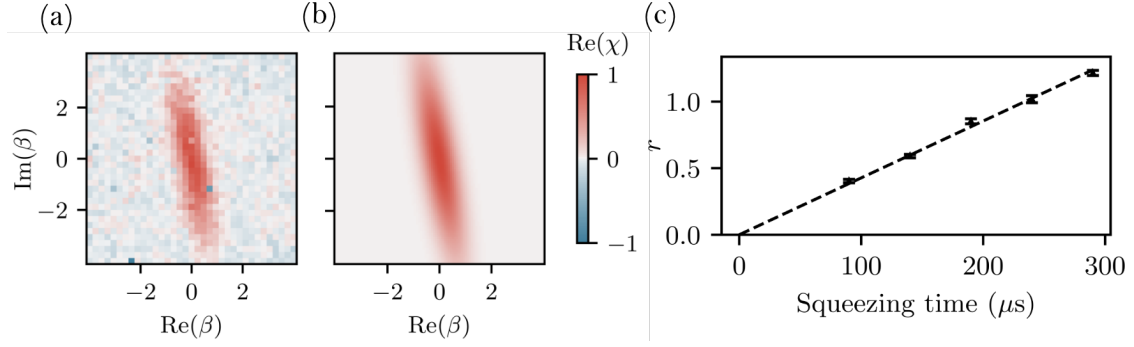


Figure 1.3: Figure: Probe SDF calibration

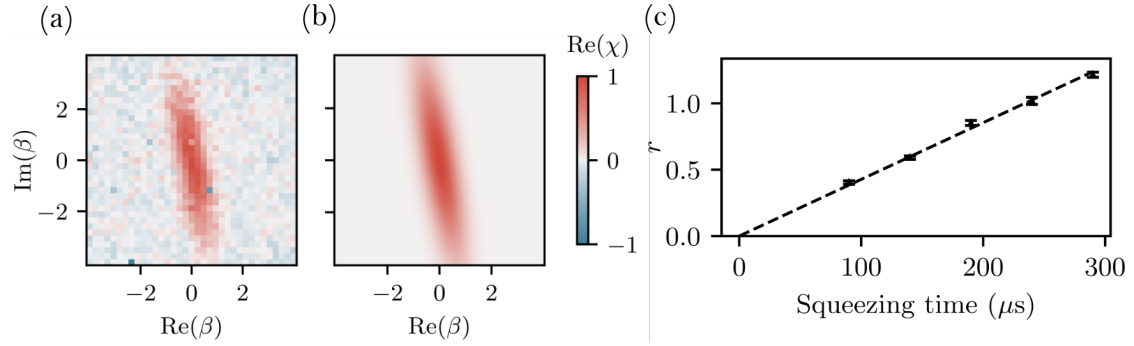


Figure 1.4: Figure: Sqz characteristic function with varying duration

Demonstration of the squeezing interaction through the two-SDF method. (a) The characteristic function, $\text{Re}(\chi(\beta))$, of the squeezed state measured using the state tomography method given by figure 1.1. (b) The simulated state taking fit parameters inferred by equation ???. (c) The fitted squeezing parameter, r , as a function of two-SDF pulse duration.

1.1.3 Outlook

- Some next steps for demonstrating full CV control on > 1 qubit and on > 1 mode.

1.2 Virtual Distillation

Short summary of section Here we develop experimental tools for demonstrating VD on the discrete variable portion of the full hybrid system.

1.2.1 Error Mitigation

Noise coupled to a quantum system results in unintended coherent, and incoherent evolution. Deviations of the realised state from the target state are known as

errors, and, if left unchecked, lead to the failure of quantum algorithms. This failure can either be due to a reduction of signal to noise ratio, or a bias in the measured expectation value, resulting in misleading conclusions. Ultimately, the goal of quantum error correction, QEC, is to detect and correct these errors through embedding information in a larger Hilbert space [**qec_review**]. Full QEC will allow for the execution of arbitrarily long algorithms ensuring errors (of up to a certain distance) do not accumulate over time. However, current QEC methods require large overheads in both qubit number and circuit depth, and are therefore not feasible on current quantum hardware. Furthermore, QEC is formulated for digital “gate-based” quantum computers, which although promise to be universal, pay for that privilege in often inefficient decompositions of target unitaries into a discrete set of gates. This poses an opportunity for near term devices to show quantum advantage using either short depth, uncorrected circuits, or analog quantum simulators, which although not universal, can provide meaningful results for specific problems [**nisq_review**]. Particularly, phrasing the problem in terms of expectation values of an observable on a physical final state, rather than a sequence of measurements on a set of logical state, allows for a more efficient use of resources. As a tactile example, consider the problem of finding the ground state energy of a Hamiltonian. This can be done by preparing a trial state, and directly measuring the expectation value of the Hamiltonian on that state, building up statistics over many samples [**variational_qc_review**]. Increasing the number of samples will reduce the statistical error, however, if the trial state is not prepared perfectly due to either environmental noise, or to imperfect application of unitary evolutions, the measured expectation value will be biased. To this end, quantum error mitigation (QEM) techniques have been developed to reduce the bias of measured expectation value of an observable at the expense of requiring more samples. Reference [8] defines QEM as “... algorithmic schemes that reduce the noise-induced bias in the expectation value by postprocessing outputs from an ensemble of circuit runs, using circuits at the same noise level as the original unmitigated circuit or above.” This highlights the crux of how information on the system is collected by a QEM

implementation, which I shall explore with the following thought experiment on the “perfect” QEM algorithm. Consider the problem of measuring the expectation value of an observable $\hat{O}$ on the target state ρ_{id} . The state is prepared by a quantum circuit, which due to noise, instead prepares the mixed state $\rho = (1 - \lambda)\rho_{id} + \lambda\rho_{err}$, where ρ_{err} is the error state. Each measurement of the observable $\hat{O}$ on the noisy state ρ will return a value o_i , and the average of these measurements will converge to the expectation value $\langle \hat{O} \rangle = \text{Tr}[\hat{O}\rho]$. Given S total samples, the final result will be,

$$\langle \hat{O} \rangle = \frac{1}{S} \left(\sum_{i=1}^{S_{id}} o_i + \sum_{j=1}^{S_{err}} o_j \right),$$

where S_{id} and S_{err} are the number of samples that return a measurement outcome from the ideal and error states respectively, with $S = S_{id} + S_{err}$. The bias in the measured expectation value from the ideal is then given by contributions from the samples S_{err} . An ideal QEM algorithm would flag these error samples, and remove them from the final average, resulting in a bias free expectation value using the restricted samples S_{id} .

- Error mitigation motivation. Mainly on contrasting to quantum error correction.
- QEM is framed as finding expectation value of observable, rather than maintaining coherence for a long computation.
- How QEM reduces bias (at the expense of required samples). Look at Cai paper for structured argument.

1.2.2 Virtual Distillation Protocol

Purification

Estimate expectation value of observable on a state without ever physically preparing the state. Purification here refers to approaching the dominant pure state of a mixed state. The k th degree purified state is given by,

$$\rho_{\text{corrected}}^{(k)} = \frac{\rho^k}{\text{Tr}[\rho^k]} = \sum_{i=1}^d \frac{\lambda_i}{\sum_{i=1}^d \lambda_i^k} |\psi_i\rangle \langle \psi_i|, \quad (1.3)$$

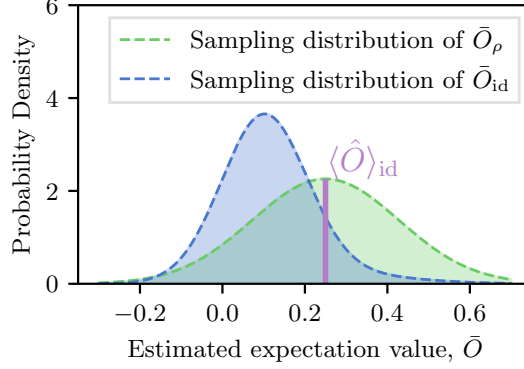


Figure 1.5: Figure: unmitigated vs mitigated expectation value, showing how bias is reduced at the expense of more samples. $\rho = 0.1\rho_{\text{id}} + 0.9\rho_{\text{err}}$. Blue curve is PDF for N shots, green is PDF for ideal shots, with $N_{\text{id}} = 0.1N$.

A schematic explanation of the goal of quantum error mitigation. Deviations of the measured expectation value of an observable $\hat{O}$ on a noisy state, ρ , and on the ideal state ρ_{id} may be separated into components originating from shot noise, and from an underlying bias. Shot noise sets the variance of the distribution of estimated expectation values, while the underlying bias shifts the center of the distribution. Error mitigation reduces bias at the cost of increasing variance. Here we demonstrate this by the ideal QEM method introduced in section 1.2.1. The underlying noisy state is given by $\rho = (1 - p)\rho_{\text{id}} + p\rho_{\text{err}}$. N shots are taken measuring $\text{Tr}(\hat{O}\rho)$. The ideal QEM flags and discards shots originating from ρ_{err} , leaving $(1 - p)N$ shots of the ideal state to estimate $\langle \hat{O} \rangle$.

where d is the dimension of the Hilbert space, λ_i are the eigenvalues of ρ , and $|\psi_i\rangle$ are the corresponding eigenvectors. If we assume that one eigenvalue is dominant, $\lambda_1 \gg \lambda_{i>1}$, then the normalised purified state exponentially approaches the dominant eigenvector, $\rho_{\text{corrected}}^{(k)} \rightarrow |\psi_1\rangle \langle \psi_1|$, as k increases. The hope, which we shall verify in section XX on coherent mismatch, is that the dominant eigenvector of the noisy state closely approximates the ideal state, and therefore effective measurements on the purified state yield accurate expectation values of the ideal state. The estimated expectation value of measuring an observable, $\hat{O}$, on this purified state is then,

$$\langle \hat{O} \rangle_{\text{corrected}} = \text{Tr}[\hat{O}\rho_{\text{corrected}}^{(k)}] = \frac{\text{Tr}[\hat{O}\rho^k]}{\text{Tr}[\rho^k]}. \quad (1.4)$$

Next, how to access this corrected state on which to make these measurements? Kozcor and Huggins show that this can be accessed without explicit preparation of the purified state. Instead the numerator and denominator may be found through preparing k copies of the noisy state, applying some derangement operation to

entangle all copies together, and measuring the desired observable on one of the copies. This can be understood through the following identity,

$$\text{Tr}[\hat{O}\rho^k] = \text{Tr}[\hat{O}^i S^{(k)} \rho^{\otimes k}], \quad (1.5)$$

where $S^{(k)}$ is the derangement operator, which permutes the k copies of the state, and $\hat{O}^i$ is the observable acting on the i th copy. We shall define the derangement operator explicitly as the cyclic permutation of k objects,

$$S^{(k)} |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_k\rangle = |\psi_2\rangle \otimes |\psi_3\rangle \otimes \dots \otimes |\psi_k\rangle \otimes |\psi_1\rangle. \quad (1.6)$$

Huggins provides a proof of this identity, which I reproduce here for the simple example of $k = 2$. The derangement operator for two copies is simply the swap operator, $S^{(2)} = \text{SWAP}$, and we will assume the observable acts only on the first copy, giving $\hat{O}^1 = \hat{O} \otimes \mathbb{I}$. The identity is then,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (1.7)$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \langle i_1 | \hat{O} | k_1 \rangle \delta_{i_2, k_2} \langle k_1, k_2 | S^{(2)} | j_1, j_2 \rangle \langle j_1 | \rho | i_1 \rangle \langle j_2 | \rho | i_2 \rangle \quad (1.8)$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \hat{O}_{i_1, k_1} \delta_{i_2, k_2} \delta_{k_1, j_2} \delta_{k_2, j_1} \rho_{j_1, i_1} \rho_{j_2, i_2} \quad (1.9)$$

and contracting the Kronecker deltas, we find,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (1.10)$$

$$= \sum_{i_1, j_1, j_1} \hat{O}_{i_1, j_2} \rho_{j_1, i_1} \rho_{j_2, j_1} \quad (1.11)$$

$$= \text{Tr}[\hat{O} \rho^2], \quad (1.12)$$

with the last equality following from the definition of matrix multiplication.

The denominator of the corrected expectation value is found by replacing the observable with the identity and measuring the expectation value of the derangement operator on k copies of the noisy state.

Access to this corrected expectation value is then possible if we can measure the multi-qubit observable $\hat{O}^i S^{(k)}$ on k copies of the noisy state. We will discuss

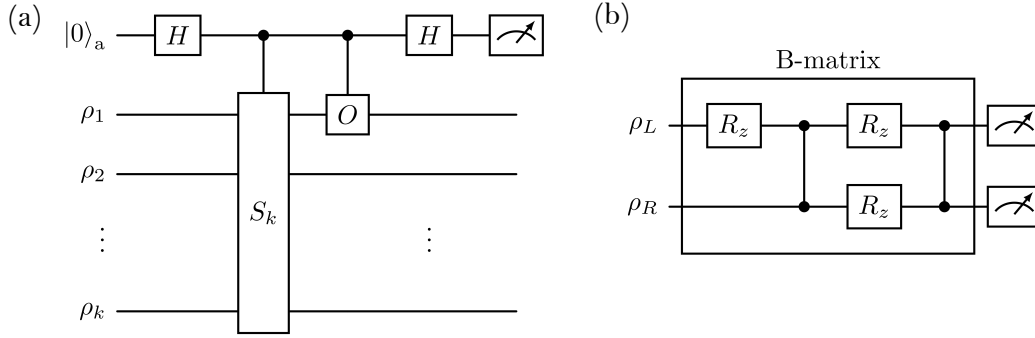


Figure 1.6: Figure: circuit diagram for ancilla based virtual distillation, Hadamard test, derangement operator by CSwap Two circuits for implementing virtual distillation. (a) Hadamard test method using controlled derangement, S_k , over k copies, and controlled observable, $\hat{O}$, applied on one copy. The ancilla qubit is measured to extract $\text{Tr}(\hat{O}\rho^k)$. Repeating the circuit without the controlled observable measures normalisation factor, $\text{Tr}(\rho^k)$, used to recover the virtually purified result, $\langle \hat{O} \rangle_{\text{VD}}$. (b) Two-copy implementation using a B -circuit, here decomposed into two entangling gates and single qubit phase gates. Measurement of both qubits in the computational basis, and classical postprocessing recovers $\text{Tr}(Z\rho^2)$ and purity, $\text{Tr}(\rho^2)$, used to recover $\langle Z \rangle_{\text{VD}}$.

two methods of implementing this measurement. First, a Hadamard test, given in Fig. ??, which uses an ancilla qubit to control the derangement and desired observable operation, and then subsequent measurement of the ancilla only. The second method, given in Fig. ??, uses a B -matrix to implement the derangement operation, and then measures the observable on one of the copies. The Hadamard test is more general, as it can be used to measure any observable, however, it requires an additional ancilla qubit and controlled operations. The B -matrix method has better qubit number and operation depth efficiency, however, requires simultaneous diagonalisation of $S^{(k)}$ and $\hat{O}^i S^{(k)}$, which is not always possible for an arbitrary observable and derangement operator.

Next, need to derive scaling of bias with copy number. And finally, discuss the variance of measurement with shot number (and how this scales with copy number?).

Circuit

Ancilla method

B-matrix protocol

If we can diagonalise both S and SO . Can apply transversally

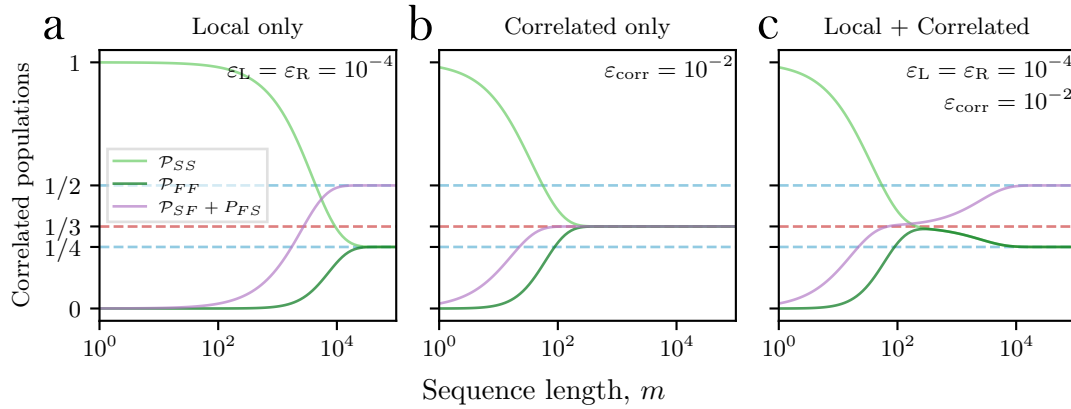


Figure 1.7: Figure: pi-4 MS gate calibration. Figure: Heuristic signal of B-matrix working. Tomography would be best!

On hybrid systems

It can be used for both spin and bosonic objects

1.2.3 Experimental Demonstration

Short description of following results.

Implementation

Protocol, circuit diagram, hardware involved.

B-matrix Calibration

pi-4 gates, single addressing

Injecting Spatial Correlations

two copy, B matrix demonstration

Analysis of this data.

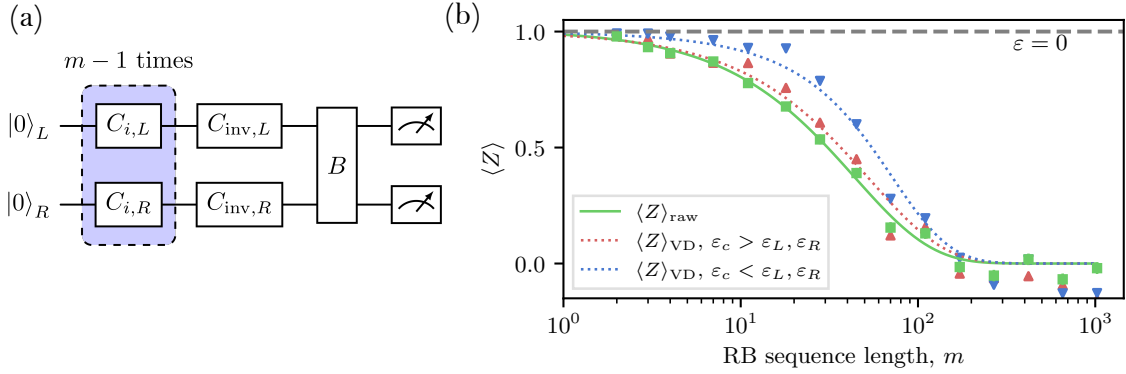


Figure 1.8: Figure: Circuit diagram of RB + VD, exp data for noise floor for two copy B-matrix, with and without spatial correlations

Verifying the performance of virtual distillation under either dominant local depolarising or correlated errors. Structured errors were injected through a fixed frequency detuning of the 729-nm laser, and the application of a sequence of random Clifford gates. In the case that identical Clifford gates, $C_{i,L} = C_{i,R}$, are applied on both qubits, the detuning error is twirled to a *common noise* channel, while in the case that Clifford gates are sampled independently for both copies, the detuning is twirled to local depolarisation channels. The local and correlated injected error-per-Clifford for both regimes are quantified using k -copy RB. Panel (a) shows the circuit of injected errors followed by the B -circuit. Panel (b) shows the effective reduction in error-per-Clifford by applying the virtual distillation circuit. The green points correspond to the raw single-qubit decay due to the physical errors, the solid green line is a fit of this decay to measure a total error-per-Clifford. The red and blue points correspond to the error mitigated Z observables for the dominant correlated errors and local depolarising errors respectively. The dotted lines correspond to the expected error mitigated single qubit decays inferred by the model given in equation ??, and the error-per-Cliffords measured by k -copy RB.

1.3 Proposed experiments and technical requirements

How CVDV and VD combine.

Extending to > 2 copies

plan for c-swap implementation.

What is difficult, what numbers we need (gate fidelities, shot number)

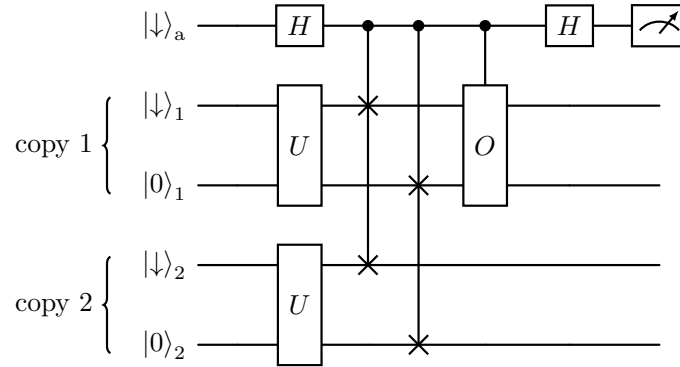


Figure 1.9: Figure: Fredkin gate on oscillator, and how to compose derangement transversally. Proposed circuit for applying virtual distillation to a hybrid system. Using the ancilla method for measuring $\text{Tr}(\hat{O}\rho^k)$, we may decompose the controlled derangement to the transversal application of Fredkin gates.

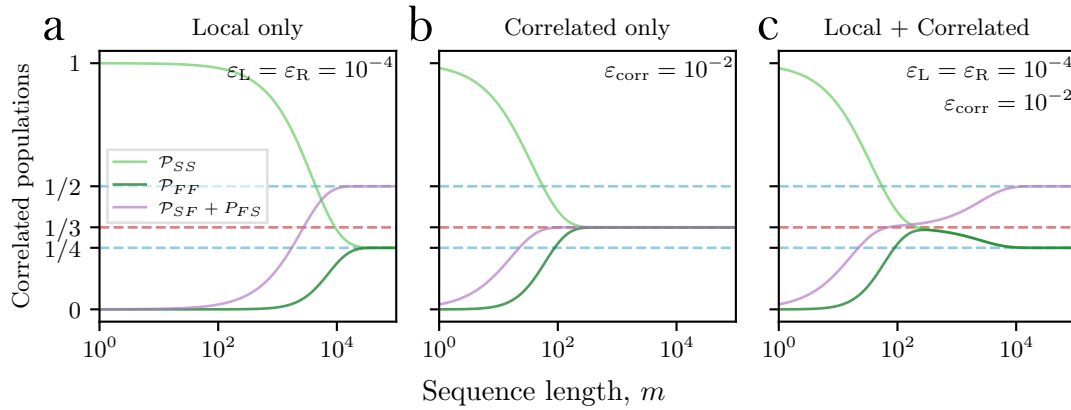


Figure 1.10: Figure: Simulation of tri-sqz state with VD to show negativity!

Virtual Distillation on Hybrid Systems

Cite Quantum virtual cooling

Why important to develop

What errors does this mitigate?

How it looks on a thermal state and on various quantum states (sqz, coherent, fock)

What is difficult, what numbers we need (gate fidelities, shot number)

1.4 Outlook

Broader perspective of these QEM research directions, and also how we can play with our larger CVDV system Can generate sqz using SW

Appendices

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